

# PDF (Probability Density Function)

A **PDF** describes how **likely** a **continuous random variable** is to take values **within a range**.

Key idea:

- It applies to **continuous** data (not discrete)
- It does **NOT** give probability at a single point
- Probability comes from the **area under the curve**

## Key Points about PDF Formula

- *If we differentiate CDF, we get the PDF of the random variable.*

$$f(y) = \frac{d}{dy} [\mathbf{F}(y)]$$

- *If we integrate PDF, we get the CDF of the random variable.*

$$F(y) = \int_{-\infty}^y f(t) dt$$

## Formal definition

For a continuous random variable (  $X$  ), a function (  $f(x)$  ) is a PDF if

1.  $f(x) \geq 0$  for all  $x$
2. Total area under the curve is 1

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

3. Probability over an interval:

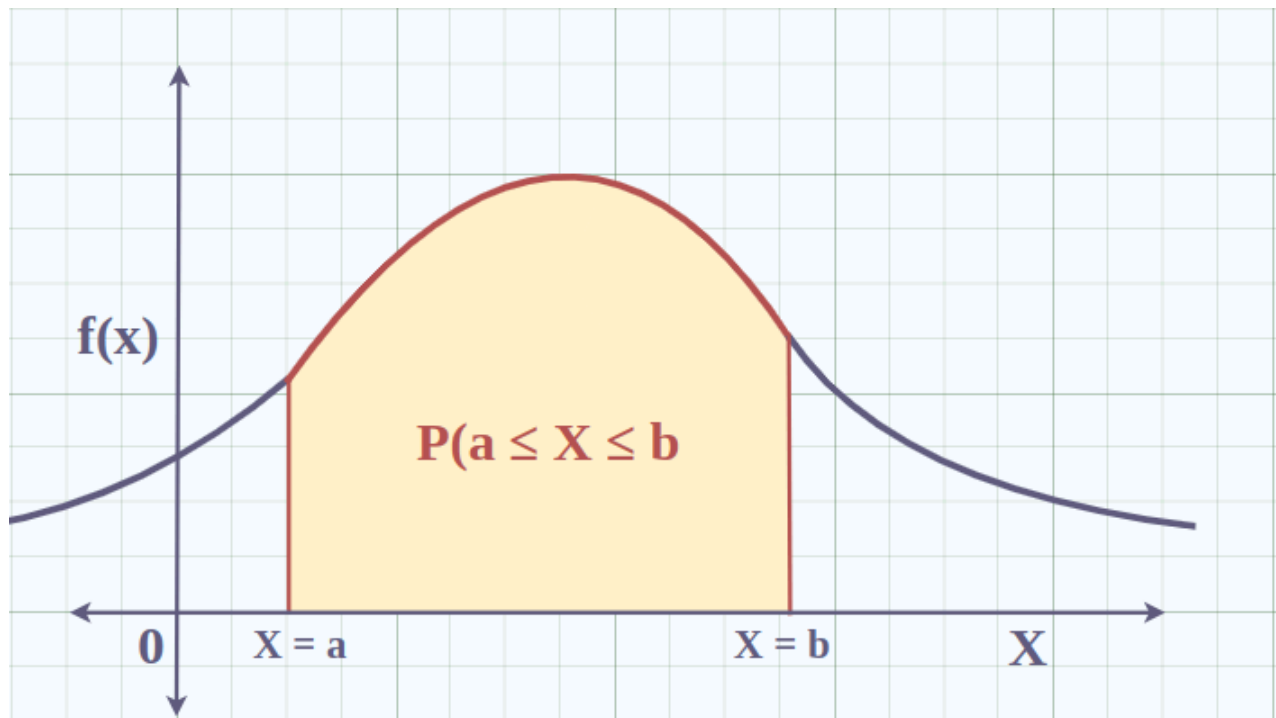
$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

## Important

(  $P(X = c) = 0$  ) for any continuous random variable

Why?

- A single point has **zero width**
- Probability only exists over **intervals**



### Simple example: Uniform distribution

Suppose:

- $X \sim \text{Uniform}(0, 10)$

PDF:

$$f(x) = \begin{cases} \frac{1}{10}, & 0 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

### Probability example

Find  $P(2 \leq X \leq 5)$ :

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{10} dx = \frac{3}{10}$$

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## Classic example: Normal distribution

If

$$X \sim N(\mu, \sigma^2)$$

PDF:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Used for:

- Heights
- Test scores
- Measurement errors

Smooth, bell-shaped curve 🔔

## PDF vs PMF

| FEATURE                  | PDF                 | PMF               |
|--------------------------|---------------------|-------------------|
| Data type                | Continuous          | Discrete          |
| Used for                 | Normal, Exponential | Binomial, Poisson |
| Single value probability | Always 0            | Can be $> 0$      |
| Probability comes from   | Area                | Sum               |

## Intuition (plain English)

- PDF height = **density**, not probability
- Probability = **area under the curve**
- Taller curve  $\neq$  more likely at a point
- Wider interval = more probability